FPGA-Based Sign Language Recognition System

FSM Diagram & Hardware Module Descriptions

CEN466 / EEN466 Advanced Digital Systems Design

1. FSM Diagram

The finite state machine (FSM) is used to explain the global behaviour of the finger-spelling recognition system based on FPGA. The FPGA repeatedly scans the five flex sensors and decodes the resulting 5-bit pattern to see which alphabet letter corresponds. The displayed first line of the LCD displays the detected letter and its binary representation. The user presses KEY0 and the current letter is stored in memory and added to the message printed on the second line of the LCD. Pressing KEY1 erases the message stored and resets the display. The system goes back to monitoring the sensor inputs after each update, enabling continuous real-time operation.

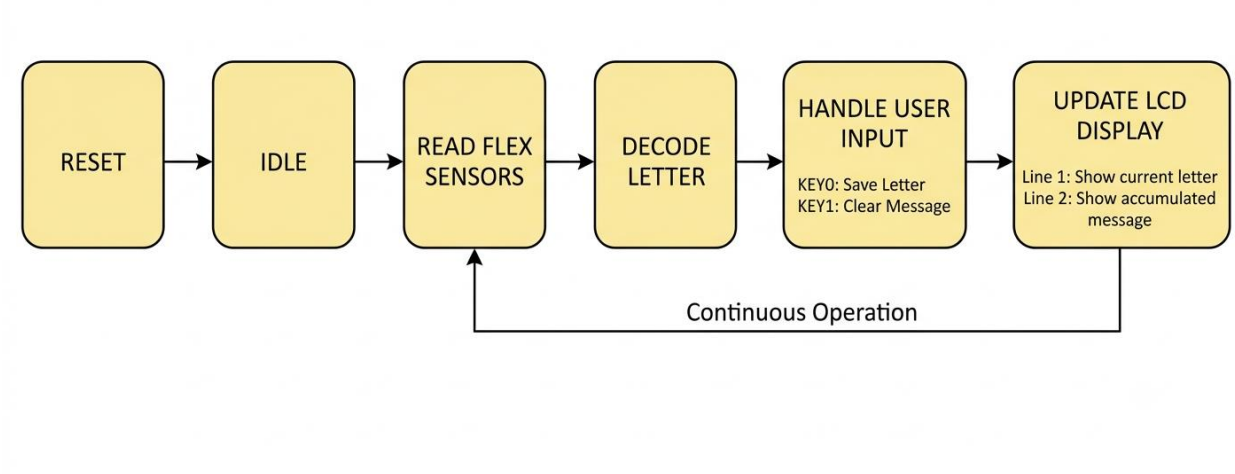


Figure 1: System-level FSM of the FPGA-based sign language recognition system

State Descriptions

Upon reset the system goes to the state IDLE and waits for the clock before it starts normal operation. Five digital input are continuously read by the FPGA from the flex sensor interface circuits. These inputs are a 5-bit pattern that indicates the finger position at the time. The pattern is fed to the VHDL logic and a corresponding alphabetical letter is mapped. The detected letter and the value of the letter in binary is shown on the first line of the LCD. On pressing KEY0, the current letter detected is stored in an internal memory buffer and added to the message that is shown in the second line of the LCD screen. Clearing Stored message and resetting buffer with the push of KEY1. The system then keeps on reading the inputs from the sensors and updates the LCD dynamically.

2. Hardware Module Descriptions

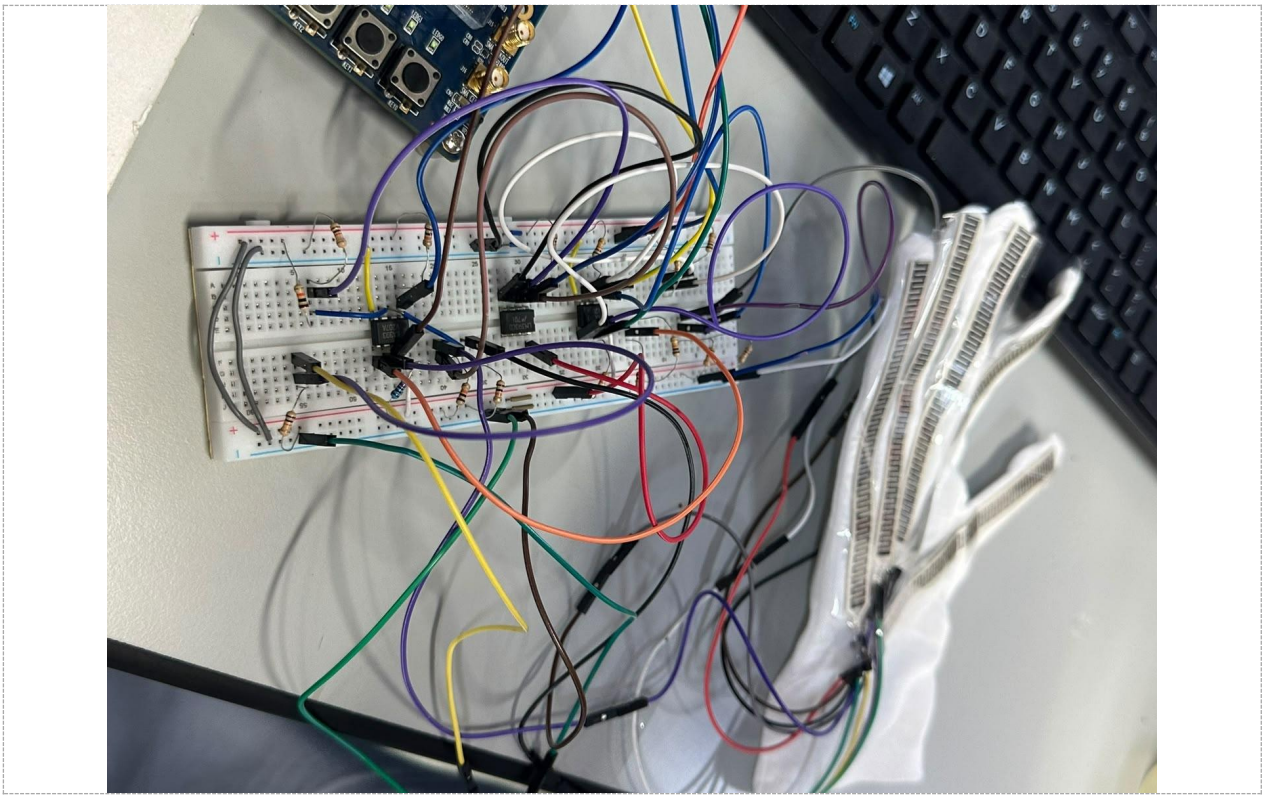


Figure 2: Comparator circuit and breadboard wiring

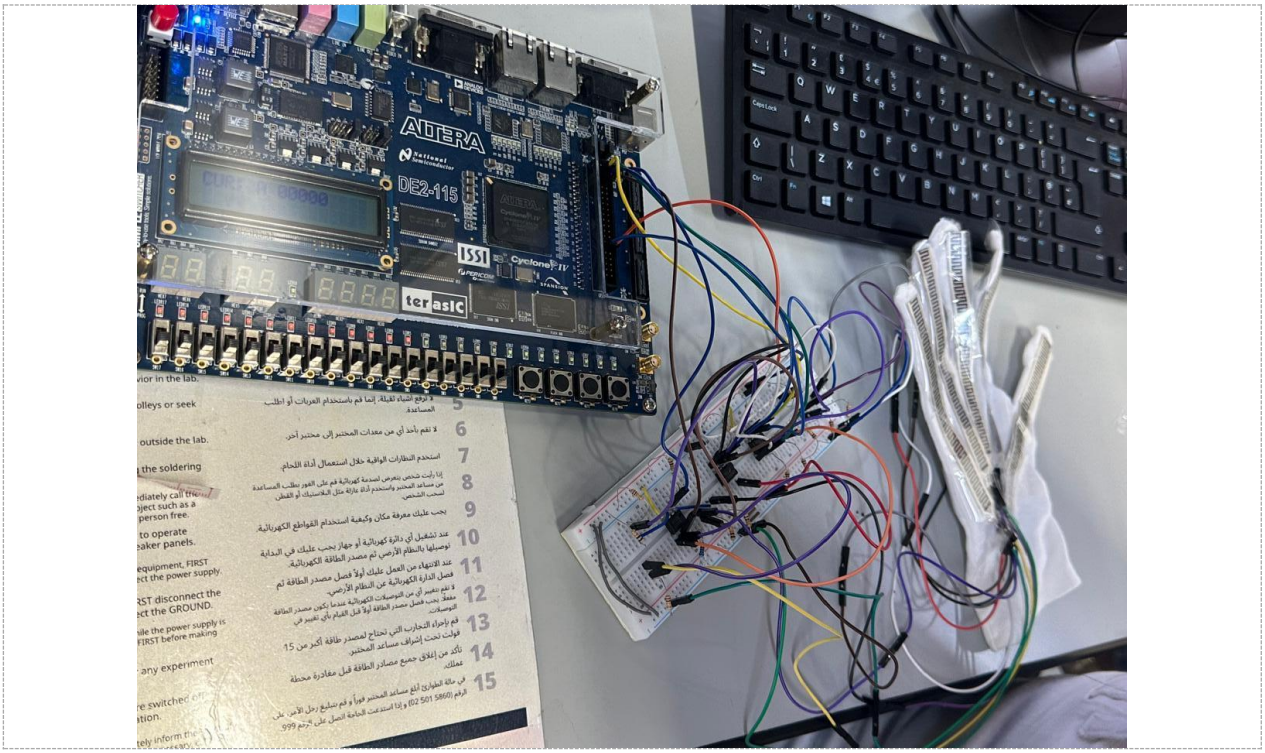


Figure 3: Full system showing DE2-115 FPGA, sensor glove, and LCD display

2.1 Flex Sensor Module

Five flex sensors (one per finger) are used in the system. A flex sensor is a resistor that changes resistance with the bending of the finger. A voltage divider and comparator circuit is connected to each sensor, and a digital signal is outputted which represents the bending state of each finger. The five sensors produce a 5-bit input pattern which is applied for alphabet recognition.

2.2 Comparator Module

To read the analog voltages produced by the flex sensors directly by the FPGA, comparator circuits are employed to convert the analog voltages to digital logic signals. A comparator compares the voltage from the sensor to a reference voltage and produces a logic HIGH or logic LOW. It means that the digital inputs are stable and it makes the FPGA processing easier since there is no need to convert from analog to digital.

2.3 DE2-115 FPGA Module

DE2-115 development board has a Cyclone IV FPGA chip that is clocked at 50 MHz. The FPGA reads the five sensor inputs all the time and implements the VHDL code that performs the alphabet recognition, push-button processing, character storage and LCD control. When testing the LEDs on the board indicate the success of the sensor states and confirm the proper functioning of the hardware.

2.4 LCD Display Module

The DE2-115 board has a built-in 16×2 LCD with a built-in controller that is compatible with the HD44780. First line of LCD displays the detected alphabet symbol and its associated 5-bit pattern. The second line shows the order in which you will store letters in it, and you can form complete words or short messages. A finite state machine (FSM) is implemented in VHDL and updates the LCD continuously to give the stable real-time feedback.